

Erdős Problem #973

EP-973

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Does there exist a constant $C > 1$ such that, for every $n \geq 2$, there exists a sequence $z_i \in \mathbb{C}$ with $z_1 = 1$ and $|z_i| \geq 1$ for all $1 \leq i \leq n$ with $\max_{2 \leq k \leq n+1} \left| \sum_{1 \leq i \leq n} z_i^k \right| < C^{-n}$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>